

GUIDELINES FOR WESTERN PROVINCE WETLANDS ZONING & RELEVANT REGULATIONS

**FOR APPLICATION IN URBAN DEVELOPMENT PLAN PREPARATION
2018**

**Prepared by the Environment & Landscape Division in consultation with
other Divisions of the Urban Development Authority (UDA) and
all other Stakeholders**

I INTRODUCTION

Approximately **20 % of the Western Province** is composed of wetlands as defined in the Ramsar Convention “areas of marsh, fen, peatland or water, where natural or artificial, permanent or temporary, with water that is static or flowing, fresh, brackish or salt, including areas of marine water the depth of which at low tide does not exceed six metres” (Article 1.1 of Ramsar Convention) with possible inclusion of “riparian and coastal zones adjacent to the wetlands, and islands or bodies of marine water deeper than six metres at low tide within the wetlands” (Article 2.1 of Ramsar Convention). **The region's wetland network is the macroscale natural drainage network affording vital stormwater drainage and flood detention, thus protecting high land from flood and making it available for intensive urban development at little or no cost. Most wetlands also serve as "green lungs" and as essential open spaces visually, spatially and climate-wise, for human health (both physical and mental), comfort and recreation. Many wetlands are high in biodiversity and some of them are significant biodiversity reserves.**

However, in view of the large percentage of wetlands in the Western Province and the lack of infrastructure serving all high land available for development, some filling for development purposes has to be permitted. This could include use of wetlands for carefully planned and implemented public infrastructure projects such as highway development, solid waste management projects and sewage disposal projects, where more suitable land is not available.

Nevertheless, it is recognized that most of the wetlands should be conserved and some preserved in order to ensure that the expected urban development in the province is sustainable. This is crucially important in view of flood disaster prevention, considering the tropical monsoon climate of the region and its vulnerability to both the south-west and north-east monsoons as well as intermonsoonal rains and future increased extreme storm events to be expected because of climate change.

The wetlands in the western province could be classified as follows:

- I. Deep freshwater marshes and swamps
- II. Shallow freshwater marsh including some abandoned clay pits and similar abandoned mining areas.
- III. Recently abandoned paddy fields
- IV. Paddy fields
- V. Other wetland agriculture areas such as reed/rush fields, low-lying grasslands (often used as grazing grounds) "Keera-Kola" fields & certain "Ovita" areas/ Madiththa /Asliyadda/Deniya
- VI. Water bodies (such as estuaries, lagoons, lakes and some abandoned clay pits) and waterways, including sea-grass beds.
- VII. Mangrove areas.

The Goal of Wetland Zoning in the Western Province is to facilitate environmentally & economically sustainable use of wetlands considering their intrinsic potential, their ecology and the needs of the surrounding population, while maintaining essential flood storage capacity and other environmental benefits.

These guidelines should be applied to all Urban Local Authority areas in the Western Province.

II ZONING

A series of workshops both within the UDA and interagency, the latter involving the Ministry of Environment, Ministry of Urban Development and all other related Government and Non Government Agencies, led to **definition of 4 basic zones:**

1. **Wetland Protection Zone.**
2. **Wetland Nature Conservation Zone.**
3. **Wetland Agriculture Zone.**
4. **Special Paddy Cultivation Zone.**

Already declared wetland wildlife sanctuaries, “Environmental Protection Areas” and areas of potential for declaration as such reserves should be zoned as **“Wetland Protection Zone”**, taking into consideration their present physical condition: in this zone, very strict regulations would apply. Land in the zone should be used predominantly for nature-based recreation and ecotourism.

Most of the other deep marsh areas, areas of known significant bio-diversity value and critical flood detention areas should be zoned as **"Wetland Nature Conservation Zone"**. This zone will also cater for nature-based tourism and recreation to a great extent, while selected areas may be suitable for ecotourism and some patches within the zone may be designated as dry-weather parks or playgrounds. Adventure/Theme Parks and the like could be suitable, provided that the wetland habitats are conserved or restored.

Areas where agriculture is of crucial importance in relation to the overall vision for Western Province development should be zoned as the **“Wetland Agriculture Zone”**. More detailed zoning could be proposed according to the nature of the particular wetland soils, topography and other agro-ecological factors in different parts of urban development areas under consideration. Dry-weather public parks and playgrounds should be allowed where essential.

Areas where it is necessary to ensure that paddy cultivation continues should be zoned as the **“Special Paddy Cultivation Zone”**.

The environmental **“precautionary principle”** should be applied in identifying such zones, until more detailed scientific studies are available. The flood detention areas specified in the JICA-SLLR & DC Stormwater Drainage Study (March 2003) should not always be considered adequate, since the

latter was a preliminary study, the 50 year design storm was used only for macro-planning and the proposed plans for the four basins addressed advocated much channel improvement and similar structural measures (which may only be implemented in the long term) in addition to the flood detention areas proposed. Therefore many areas would be categorized as “Wetland Nature Conservation Zone” or “Wetland Agriculture Zone”.

The remaining areas may be designated as the “**Low-lying Potential Development Zone**”, where development of various types could be permitted. Certain wetland patches/strips (or groups of such areas) close to town centers where public development is needed and suitable high land is lacking, could be in this zone. Small isolated wetland patches and tips of wetland strips may also be zoned as Low-lying Potential Development Zone, provided drainage and flood prevention requirements are satisfied. Wetlands composed of abandoned clay pits and other pits/depressions/quarries where open-cast mining or quarrying has occurred, could also be in this zone. Conditions of development could be site specific according to whether the site was a highland or wetland before mining and the current needs/situation in the area.

Specific locations for public outdoor recreation spaces of different categories (including dry-weather playgrounds) in accordance with the UDA’s “hierarchy of parks” and “nesting” spatial concepts for such services and their service areas, should be identified in each local authority area as a component of the urban development plan. Many (especially where high land cannot be made available) will have to be located in wetlands and some of these could still serve as flood detention areas. Recreation-conducive physical characteristics of the locality should also be taken into account in identifying these locations.

III GENERAL CONDITIONS AND GUIDELINES FOR ALL FOUR BASIC WETLAND ZONES

CONDITIONS

- Flood storage capacity shall be maintained as specified by the Sri Lanka Land Reclamation & Development Corporation (SLLR & DC) in co-ordination with the Irrigation Department (ID), the Central Environmental Authority (CEA), the UDA, the Wild Life Conservation Department (WLCD), the (Forest Department (FD) and the Agrarian Development Department (ADD). Increased storm water run-off owing to increase in built-up areas and owing to expected extreme climatic events as a result of global warming and climate change shall be taken into account.
- The environmental clearance or environmental protection licence obtained for any development from the CEA or the Local Authority shall be renewed annually.
- The services of relevant qualified professionals such as hydrologists, ecologists, town planners, environmental engineers, landscape architects, architects, economists and sociologists, shall be obtained by developers for planning, design and supervision, as necessary. All buildings shall be designed by chartered architects.
- Clearance shall be obtained from the following agencies prior to development of any low-lying lands as indicated in Circular No. 13 dated 29th September 1990, issued by the Director General of UDA: UDA, SLLR &DC, Agrarian Development Department, (if the site was/is in agricultural use), relevant local authority, CEA. Where necessary, clearance of the Wildlife Conservation Department, Irrigation Department (especially for Attanagalla Oya basin, Kelani Ganga Basin and Kalu Ganga Basin), Coast Conservation Coastal Resources Management Department (CC&CMD), Forest Department and National Building Research Organization shall also be obtained.
- Natural mangrove areas and lagoons shall be preserved as they are breeding grounds for fisheries.

- Environmentally friendly solid waste, sewage and wastewater management systems shall be planned and put into practice.
- In general there shall not be any construction within water bodies and waterways, but exceptions may be made for focal features/buildings, piers, picnic shelters, cabanas on stilts, fishing decks, boardwalks, etc. in keeping with a project masterplan or design guide plan approved by the UDA, not compromising drainage or flood detention capacity.
- Areas of special ecological interest shall be preserved and in such areas no vegetation or animals shall be removed (except intrusive species such as "Vel artha").
- In general, waste dumping shall not be permitted.
- Permission should be obtained from Coast Conservation & Coastal Resources Management Department for any prior development of low-lying lands within "Coastal Zone".

GUIDELINES

- In any areas permitted for filling, sustainable stormwater drainage systems should be used, integrating on-site storage, infiltration, soakaways and aquifer recharge, while minimizing downstream discharge and preventing increased upstream flooding.
- Areas of outstanding landscape/cultural/historical value should be conserved.
- Legal public footpaths and public bathing wells should be conserved or replaced in suitable nearby locations.
- Use of wetlands for water transport and/or water supply should be integrated as far as possible.
- Building design should aim for energy efficiency and use natural ventilation and solar energy as far as possible.

WESTERN PROVINCE WETLAND ZONING REGULATIONS

ZONE – SPECIFIC DEVELOPMENT CONDITIONS

Conditions of Development	Wetland Protection Zone	Wetland Nature Conservation Zone	Wetland Agriculture Zone	Special Paddy Cultivation Zone
a) Flood storage capacity.	shall be maintained in accordance with a Master Plan approved by the SLLR & DC and where relevant, the ID as well			
b) Minimum plot size for the purpose of any building construction.	4 ha (10 acres) In smaller plots the permitted uses are the same but no buildings shall be allowed.	2 ha (5 acres) In smaller plots the permitted uses are the same but no buildings shall be allowed.	1 ha (2.5 acres) In smaller plots the permitted uses are the same but no buildings shall be allowed.	Not relevant as buildings not permitted.
c) Maximum area where filling permitted.	2% of the site area (if needed for roads, vehicle parking, toilets and sewage disposal/treatment required for site management).	5% of the site area (if needed for roads, vehicle parking, toilets and sewage disposal/treatment required for site management).	10% of the site area, or as determined by a stormwater management master plan approved by SLLR & DC, whichever is less.	No filling permitted.

Conditions of Development	Wetland Protection Zone	Wetland Nature Conservation Zone	Wetland Agriculture Zone	Special Paddy Cultivation Zone
d) Maximum plot coverage (area covered by buildings).	1% of the total project area; all buildings on stilts excepting toilets, which may be on filled land.	3% of the total project area; all buildings on stilts excepting toilets which may be on filled land.	5% of the total project area.	Buildings not permitted.
e) Maximum area permitted for project infrastructure, i.e. Electricity/Water Supply/Telecommunication/Roads.	4% of the total project area.	7% of the total project area.	7% of the total project area.	Not permitted.
f) Maximum permitted ground floor area of an individual building.	100 sq. m.	300 sq. m.	300 sq. m.	Buildings not permitted.
g) Maximum permitted building height.	7 m from the natural ground level (except in the case of a few look-out posts/observation towers/tower hides which are compatible with the overall concept)	7 m from the natural ground level (except in the case of a few look-out posts/observation towers/tower hides which are compatible with the overall concept)	7 m from the natural or filled ground level.	Buildings not permitted.

Conditions of Development	Wetland Protection Zone	Wetland Nature Conservation Zone	Wetland Agriculture Zone	Special Paddy Cultivation Zone
h) Building type.	All buildings except toilets shall be on stilts in wetland (not filled area). They shall be isolated or in aesthetically pleasing clusters, with visually compatible, attractive “roof-scapes”. They shall not block views of the open area from adjoining public roads and other public areas and they should be environmentally friendly.		Buildings shall be isolated or in aesthetically pleasing clusters, with visually compatible, attractive “roof-scapes”. They shall not block views of the open area from adjoining public roads and other public areas and they should be environmentally friendly.	Buildings not permitted.
i) Sub-division	Not permitted	Not permitted	Not permitted	Not permitted.
j) Boundary demarcation.	Boundary walls not permitted. Visually compatible boundary fencing which does not hamper stormwater flow may be permitted only along the boundary with adjoining existing high land. Any other boundaries may be demarcated only by visually compatible boundary markers at minimum intervals of 10m. Both the above will be subject to approval of the UDA, on a site specific basis.	Boundary walls not permitted. Visually compatible boundary fencing which does not hamper stormwater flow may be permitted only along the boundary with adjoining existing high land. Any other boundaries may be demarcated only by visually compatible boundary markers at minimum intervals of 10m. Both the above will be subject to approval of the UDA, on a site specific basis.	Boundary walls not permitted. Visually compatible boundary fencing which does not hamper stormwater flow may be permitted only along the boundary with adjoining existing high land. Any other boundaries may be demarcated only by visually compatible boundary markers at minimum intervals of 10m. Both the above will be subject to approval of the UDA, on a site specific basis.	Boundary walls not permitted. Visually compatible boundary fencing which does not hamper stormwater flow may be permitted only along the boundary with adjoining existing high land. Any other boundaries may be demarcated only by visually compatible boundary markers at minimum intervals of 10m. Both the above will be subject to approval of the design by the UDA, on a site specific basis.

k) Relaxation of above conditions in exceptional cases.	Not permitted.	N.B. In the rare event of having to accommodate a vital public infrastructure project, the above conditions may be relaxed.	N.B. In the rare event of having to accommodate a vital public infrastructure project, the above conditions may be relaxed.	N.B. In the rare event of having to accommodate a vital public infrastructure project, the above conditions may be relaxed.
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N. B. In the case of “prescribed” projects under the Environmental Act, additional site-specific conditions shall be imposed by the CEA, SLLR & DC, UDA, ADD , CC& CRMD & ID, if necessary.

PERMITTED USES

Wetland Protection Zone	Wetland Nature Conservation Zone	Wetland Agriculture Zone	Special Paddy Cultivation Zone
Permitted activities should be implemented according to the already gazetted (relevant gazettes) area as a Wildlife sanctuaries, Environmental sensitive areas etc. All other activities prohibited	• Eco-friendly restaurants • Mini Conference Centers • Outdoor fitness/exercise facilities • Wetland museums • Eco-tourism facilities (i.e. small-scale, eco-friendly, tourist development such as cabanas on stilts) • Theme Parks, conserving/restoring the natural ecosystem. • Other nature based recreation activities • Open grassed areas (as public parks/playgrounds). • Traditional Fishing • Flower collection • Water-transport if carefully integrated and operated • Dairies • Wetland forestry • National infrastructure projects • Irrigation/flood protection structures. • Wetland organic agriculture. All other activities prohibited	• All uses permitted in Wetland Nature Conservation Zone. • Wetland Agriculture (including animal husbandry except piggeries, prawn farms, and poultry forms having a stock exceeding 200 birds) • Fisheries • Arboriculture, agro-forestry, fuel-wood forestry and conservation forestry including natural forest communities • Sales outlets for the produce of the plot/project • Non-polluting agricultural processing centers • Mining in accordance with (GS & MB) and CEA guidelines, conditions and regulations, including site rehabilitation and recreation (which should be for agriculture/fisheries/forestry only) • Public Parks and Playgrounds • Important Public infrastructure projects. All other activities prohibited	• Permitted uses should be implemented according to the Agrarian services Act. and all other activities are prohibited.